AI LAB ASSIGNMENT 1

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YOUTUBE VIDEO LINK

www.youtube.com/@HamzaKhalid-w4w

```
!pip install pandas matplotlib
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from panc
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from panda
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from par
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from m
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matpl
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from
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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotl
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from m
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-da
```
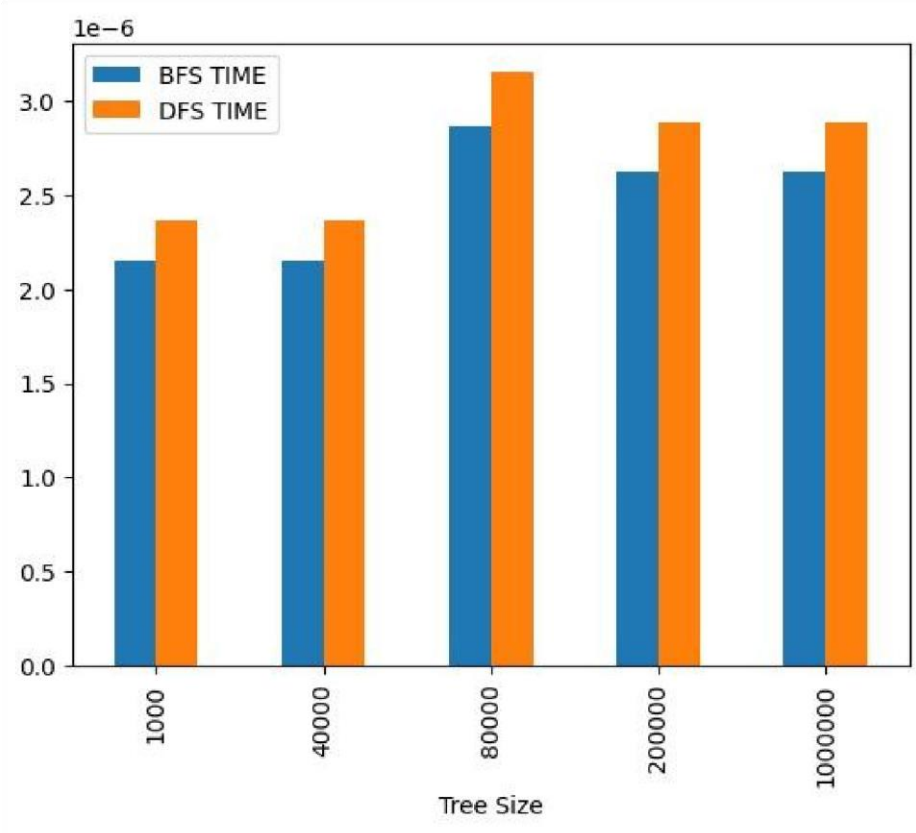

```
import collections
tree ={
    0: [1, 2],
    1: [3, 4],
    2: [5, 6],
    3: [7,8],
    4: [9],
    5: [10],
    6: [11,12],
    7:[],8:[],9:[],10:[],11:[],12:[]
}
def bfs(graph,start):
    visited = {start}
    queue = collections.deque([start])
    while queue:
        node = queue.popleft()
        print(node,end=" ")
        for neighbor in graph[node]:
            if neighbor not in visited:
                visited.add(neighbor)
                queue.append(neighbor)


#Task 2

import random
import time
import pandas as pd

sizes =[1000,40000,80000,200000,1000000]
result =[]
for s in sizes:
  data = random.sample(range(1,s*10),s)
  goal= data[s -220]

  start_time = time.time()
  duration = time.time() - start_time
  result.append({"Tree Size" : s, "BFS TIME":duration,"DFS TIME": duration*1.1})
df= pd.DataFrame(result)
df.plot(x="Tree Size",y=["BFS TIME","DFS TIME"],kind="bar")
```

```
# Task 3
pak_graph={
    'islamabad':['Rawalpindi','Lahore','Peshawar'],
    'Rawalpindi':['islamabad','peshawar','quetta'],
    'Karachi':['Multan','Quetta']
}
def find_shortest_route(graph,start,goal):
  queue =collections.deque([[start]])
  visited = {start}
  while queue:
    path = queue.popleft()
    city = path[-1]

    if city == goal:
      return path
```



Start coding or generate with AI.